MICHELE COTI ZELATI

https://micotize.github.io/

Department of Mathematics
Imperial College London
London SW7 2AZ, UK

Office: Huxley Building 6M33
Email: m.coti-zelati@imperial.ac.uk
Citizenship: Italian, British

Employment

2024–	**Professor**, Imperial College London, UK
2022–2024	**Reader**, Imperial College London, UK
2019–2022	**Senior Lecturer**, Imperial College London, UK
2017–2019	**Chapman Fellow**, Imperial College London, UK
2014–2017	**Brin Postdoctoral Fellow**, University of Maryland, College Park, MD

Education

2009–2014	**PhD in Mathematics**, Indiana University, Bloomington, IN
2006–2008	**M.S. in Mathematical Engineering**, Politecnico di Milano, Italy
2003–2006	**B.S. in Mathematical Engineering**, Politecnico di Milano, Italy

Academic Honors and Awards

2024–2027	**Royal Society** University Research Fellowship extension (URF\R1\191492)
2023–2024	**Royal Society** Enhanced Research Expenses (RF\ERE\231047)
2023–2028	**ERC Starting Grant** STABLE-CHAOS [funded by **EPSRC** due to UK non-association]
2022–2023	**Royal Society** Enhanced Research Expenses (RF\ERE\210298)
Feb. 2021	**Fellow of the Higher Education Academy**, Descriptor 2
2021–2023	**Royal Society** International Exchanges 2020 Cost Share (IEC\R2\202029)
Nov. 2020	**Abilitazione Scientifica Nazionale**, habilitation to full professor in Italy
2020	**GNAMPA** project Co-I, National Institute of Mathematics, Italy
2019–2024	**Royal Society** University Research Fellowship (URF\R1\191492)
2017–2018	**NSF Research grant** DMS-1713886
Spring 2014	**Rothrock Teaching Award**, Indiana University
Spring 2013	**Charles H. Stammer Fellowship**, Indiana University

Editorial boards

2025–	EMS Magazine
2023–	Journal of Functional Analysis
2023–	Journal of Hyperbolic Differential Equations
2022–	Proceedings of the Royal Society of Edinburgh Section A: Mathematics

Supervision and mentoring

Postdocs	Rajendra Beekie (2025–2026), Marco Bagnara (2025–), Yupei Huang (2025–), Niklas Knobel (2024–2026), Victor Navarro Fernández (2023–2026), Lucas Ertzbischoff (2023–2024), Joonhyun La (2022–2023), Michele Dolce (2020–2022)
PhD Students	Luca Melzi (2024–, w/Matias Delgadino), Haoling Xiang (2023–, w/Pierre Germain), David Villringer (2022–2026, w/Martin Hairer), Augusto Del Zotto (2020–2024), Marc Nualart Batalla (2020–2024), Michele Dolce (2018–2020)
MSc Students	Caterina Cogo (Università di Milano, 2026), Yiting Yao (2026), Averkios Averkiou (2022), Chia-Chun Lo (2022), Giordano Scarciotti (2020)
UROP Students	Leya Quan (ENS-Lyon, 2025), Yiting Yao (2024), Munazza Sarwar (2021), Chun-Hei Lam (2020)

Publications

[62] M. Coti Zelati, M. Sorella & D. Villringer. *Smooth autonomous fast dynamo action on the three-torus*. arXiv e-prints, arXiv:260904153 (Sept. 2026).

[61] M. Coti Zelati, M. Sorella & D. Villringer. *A fast dynamo on the three-torus*. arXiv e-prints, arXiv:2603.09861 (Mar. 2026).

[60] M. Coti Zelati, L. Ertzbischoff & D. Gerard-Varet. *Stability analysis for active Brownian particle models*. arXiv e-prints, arXiv:2512.17649 (Dec. 2025).

[59] M. Coti Zelati. *Invariant Measures in the Small Noise Inviscid Limit*. *Stochastics in Fluids*. Adv. Math. Fluid Mech. Birkhäuser/Springer, Cham, 2026, 53–103.

[58] M. Coti Zelati, M. Hairer & D. Villringer. *A Stochastic RAGE Theorem and Enhanced Dissipation for Transport Noise*. arXiv e-prints, arXiv:2507.11422 (July 2025).

[57] M. Coti Zelati, M. Sorella & D. Villringer. *Alpha-unstable flows and the fast dynamo problem*. arXiv e-prints, arXiv:2504.00855 (Apr. 2025).

[56] M. Coti Zelati, A. Del Zotto & K. Widmayer. *On the stability of viscous three-dimensional rotating Couette flow*. arXiv e-prints, arXiv:2501.17735 (Jan. 2025).

[55] M. Coti Zelati & V. Navarro-Fernández. *Three-dimensional exponential mixing and ideal kinematic dynamo with randomized ABC flows*. J. Dynam. Differential Equations 38.suppl. 1 (2026), 169–199.

[54] M. Coti Zelati & M. G. Delgadino. *Entropy maximization in the two-dimensional Euler equations*. Anal. PDE 19.3 (2026), 505–538.

[53] M. Coti Zelati, H. Dietert & D. Gérard-Varet. *Nonlinear stability for active suspensions*. Ann. Inst. H. Poincaré C Anal. Non Linéaire 43.5 (2026), 1145–1199.

[52] R. Bianchini, M. Coti Zelati & L. Ertzbischoff. *Ill-posedness of the hydrostatic Euler-Boussinesq equations and failure of hydrostatic limit*. Comm. Math. Phys. 406.10 (2025), Paper No. 254, 50.

[51] M. Coti Zelati, A. Del Zotto & K. Widmayer. *Stability of Viscous Three-Dimensional Stratified Couette Flow via Dispersion and Mixing*. Comm. Pure Appl. Math. 79.12 (2026), 2435–2490.

[50] G.-Q. G. Chen, M. Coti Zelati & C. C. Yeung. *Vanishing Mach Number Limit of Stochastic Compressible Flows*. arXiv e-prints, arXiv:2311.14660 (Nov. 2023).

[49] M. Coti Zelati, T. D. Drivas & R. S. Gvalani. *Mixing by statistically self-similar Gaussian random fields*. J. Stat. Phys. 191.5 (2024), Paper No. 61, 11.

[48] R. Bianchini, M. Coti Zelati & M. Dolce. *Symmetrization and asymptotic stability in non-homogeneous fluids around stratified shear flows*. *Séminaire Laurent Schwartz—Équations aux dérivées partielles et applications. Année 2022–2023*, Exp. No. V, 17.

[47] M. Coti Zelati & M. Nualart. *Limiting absorption principles and linear inviscid damping in the Euler-Boussinesq system in the periodic channel*. Comm. Math. Phys. 406.3 (2025), Paper No. 57, 106.

[46] M. Coti Zelati & M. Nualart. *Explicit solutions and linear inviscid damping in the Euler-Boussinesq equation near a stratified Couette flow in the periodic strip*. J. Hyperbolic Differ. Equ. 21.3 (2024), 659–680.

[45] M. Coti Zelati & A. Del Zotto. *Suppression of lift-up effect in the 3D Boussinesq equations around a stably stratified Couette flow*. Quart. Appl. Math. 83.2 (2025), 389–401.

[44] M. Coti Zelati, G. Crippa, G. Iyer & A. L. Mazzucato. *Mixing in incompressible flows: transport, dissipation, and their interplay*. Notices Amer. Math. Soc. 71.5 (2024), 593–604.

[43] M. Coti Zelati, M. Dolce & C.-C. Lo. *Diffusion enhancement and Taylor dispersion for rotationally symmetric flows in discs and pipes*. J. Math. Fluid Mech. 26.1 (2024), Paper No. 12, 9.

[42] E. Bruè, M. C. Zelati & E. Marconi. *Enhanced dissipation for two-dimensional Hamiltonian flows*. Arch. Ration. Mech. Anal. 248.5 (2024), Paper No. 84, 37.

[41] J. Bedrossian, M. Coti Zelati & M. Dolce. *Taylor dispersion and phase mixing in the non-cutoff Boltzmann equation on the whole space*. Proc. Lond. Math. Soc. (3) 129.1 (2024), Paper No. e12616, 70.

[40] M. Coti Zelati, H. Dietert & D. Gérard-Varet. *Orientation mixing in active suspensions*. Ann. PDE 9.2 (2023), Paper No. 20, 53.

[39] A. Blumenthal, M. Coti Zelati & R. S. Gvalani. *Exponential mixing for random dynamical systems and an example of Pierrehumbert*. Ann. Probab. 51.4 (2023), 1559–1601.

[38] M. Coti Zelati & T. Gallay. *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*. J. Lond. Math. Soc. (2) 108.4 (2023), 1358–1392.

[37] J. Bedrossian, R. Bianchini, M. Coti Zelati & M. Dolce. *Nonlinear inviscid damping and shear-buoyancy instability in the two-dimensional Boussinesq equations*. Comm. Pure Appl. Math. 76.12 (2023), 3685–3768.

[36] M. Coti Zelati, M. Dolce, Y. Feng & A. L. Mazzucato. *Global existence for the two-dimensional Kuramoto-Sivashinsky equation with a shear flow*. J. Evol. Equ. 21.4 (2021), 5079–5099.

[35] M. Colombo, M. Coti Zelati & K. Widmayer. *Mixing and diffusion for rough shear flows*. Ars Inven. Anal. (2021), Paper No. 2, 22.

[34] M. Coti Zelati & G. A. Pavliotis. *Homogenization and hypocoercivity for Fokker-Planck equations driven by weakly compressible shear flows*. IMA J. Appl. Math. 85.6 (2020), 951–979.

[33] M. Coti Zelati, T. M. Elgindi & K. Widmayer. *Stationary structures near the Kolmogorov and Poiseuille flows in the 2d Euler equations*. Arch. Ration. Mech. Anal. 247.1 (2023), Paper No. 12, 37.

[32] R. Bianchini, M. Coti Zelati & M. Dolce. *Linear inviscid damping for shear flows near Couette in the 2D stably stratified regime*. Indiana Univ. Math. J. 71.4 (2022), 1467–1504.

[31] M. Coti Zelati & M. Hairer. *A noise-induced transition in the Lorenz system*. Comm. Math. Phys. 383.3 (2021), 2243–2274.

[30] M. Coti Zelati & T. D. Drivas. *A stochastic approach to enhanced diffusion*. Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) 22.2 (2021), 811–834.

[29] M. Coti Zelati. *Stable mixing estimates in the infinite Péclet number limit*. J. Funct. Anal. 279.4 (2020), 108562, 25.

[28] M. Coti Zelati & M. Dolce. *Separation of time-scales in drift-diffusion equations on $\mathbb{R}^2$*. J. Math. Pures Appl. (9) 142 (2020), 58–75.

[27] J. Bedrossian, M. C. Zelati, S. Punshon-Smith & F. Weber. *Sufficient conditions for dual cascade flux laws in the stochastic 2d Navier-Stokes equations*. Arch. Ration. Mech. Anal. 237.1 (2020), 103–145.

[26] M. Coti Zelati, T. M. Elgindi & K. Widmayer. *Enhanced dissipation in the Navier-Stokes equations near the Poiseuille flow*. Comm. Math. Phys. 378.2 (2020), 987–1010.

[25] M. Coti Zelati, M. G. Delgadino & T. M. Elgindi. *On the relation between enhanced dissipation timescales and mixing rates*. Comm. Pure Appl. Math. 73.6 (2020), 1205–1244.

[24] J. Bedrossian, M. Coti Zelati, S. Punshon-Smith & F. Weber. *A sufficient condition for the Kolmogorov 4/5 law for stationary martingale solutions to the 3D Navier-Stokes equations*. Comm. Math. Phys. 367.3 (2019), 1045–1075.

[23] M. Coti Zelati, N. Glatt-Holtz & K. Trivisa. *Invariant measures for the stochastic one-dimensional compressible Navier-Stokes equations*. Appl. Math. Optim. 83.3 (2021), 1487–1522.

[22] M. Coti Zelati & C. Zillinger. *On degenerate circular and shear flows: the point vortex and power law circular flows*. Comm. Partial Differential Equations 44.2 (2019), 110–155.

[21] J. Bedrossian, M. Coti Zelati & V. Vicol. *Vortex axisymmetrization, inviscid damping, and vorticity depletion in the linearized 2D Euler equations*. Ann. PDE 5.1 (2019), Paper No. 4, 192.

[20] M. Coti Zelati & P. Kalita. *Smooth attractors for weak solutions of the SQG equation with critical dissipation*. Discrete Contin. Dyn. Syst. Ser. B 22.5 (2017), 1857–1873.

[19] M. Coti Zelati. *Long-time behavior and critical limit of subcritical SQG equations in scale-invariant Sobolev spaces*. J. Nonlinear Sci. 28.1 (2018), 305–335.

[18] J. Bedrossian & M. Coti Zelati. *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows*. Arch. Ration. Mech. Anal. 224.3 (2017), 1161–1204.

[17] P. Constantin, M. Coti Zelati & V. Vicol. *Uniformly attracting limit sets for the critically dissipative SQG equation*. Nonlinearity 29.2 (2016), 298–318.

[16] J. Bedrossian, M. Coti Zelati & N. Glatt-Holtz. *Invariant measures for passive scalars in the small noise inviscid limit*. Comm. Math. Phys. 348.1 (2016), 101–127.

[15] M. Coti Zelati & V. Vicol. *On the global regularity for the supercritical SQG equation*. Indiana Univ. Math. J. 65.2 (2016), 535–552.

[14] M. Coti Zelati & C. G. Gal. *Singular limits of Voigt models in fluid dynamics*. J. Math. Fluid Mech. 17.2 (2015), 233–259.

[13] M. Coti Zelati & P. Kalita. *Minimality properties of set-valued processes and their pullback attractors*. SIAM J. Math. Anal. 47.2 (2015), 1530–1561.

[12] M. Coti Zelati, A. Huang, I. Kukavica, R. Temam & M. Ziane. *The primitive equations of the atmosphere in presence of vapour saturation*. Nonlinearity 28.3 (2015), 625–668.

[11] A. Bousquet, M. Coti Zelati & R. Temam. *Phase transition models in atmospheric dynamics*. Milan J. Math. 82.1 (2014), 99–128.

[10] M. Coti Zelati, M. Frémond, R. Temam & J. Tribbia. *The equations of the atmosphere with humidity and saturation: uniqueness and physical bounds*. Phys. D 264 (2013), 49–65.

[9] M. Coti Zelati. *Remarks on the approximation of the Navier-Stokes equations via the implicit Euler scheme*. Commun. Pure Appl. Anal. 12.6 (2013), 2829–2838.

[8] M. Coti Zelati. *On the theory of global attractors and Lyapunov functionals*. Set-Valued Var. Anal. 21.1 (2013), 127–149.

[7] M. Coti Zelati, F. Dell'Oro & V. Pata. *Energy decay of type III linear thermoelastic plates with memory*. J. Math. Anal. Appl. 401.1 (2013), 357–366.

[6] M. Coti Zelati & F. Tone. *Multivalued attractors and their approximation: applications to the Navier-Stokes equations*. Numer. Math. 122.3 (2012), 421–441.

[5] M. Coti Zelati & R. Temam. *The atmospheric equation of water vapor with saturation*. Boll. Unione Mat. Ital. (9) 5.2 (2012), 309–336.

[4] M. Coti Zelati, C. Giorgi & V. Pata. *Steady states of the hinged extensible beam with external load*. Math. Models Methods Appl. Sci. 20.1 (2010), 43–58.

[3] M. Coti Zelati, V. Pata & R. Quintanilla. *Regular global attractors of type III thermoelastic extensible beams*. Chinese Ann. Math. Ser. B 31.5 (2010), 619–630.

[2] M. Conti & M. Coti Zelati. *Attractors for the Cahn-Hilliard equation with memory in 2D*. Nonlinear Anal. 72.3-4 (2010), 1668–1682.

[1] M. Coti Zelati. *Global and exponential attractors for the singularly perturbed extensible beam*. Discrete Contin. Dyn. Syst. 25.3 (2009), 1041–1060.

Schools and mini-courses

Oct. 2025	**End of CRISIS: a mathematical school/conference on fluid dynamics**, CIRM, France
Sep. 2024	**Summer school in fluid dynamics and nonlinear PDEs**, University of Padova, Italy
Aug. 2023	**Stochastics in Fluids**, Czech Academy of Sciences, Czechia
Jun. 2023	**New trends in mathematical fluid dynamics**, Université Grenoble Alpes, France
Jan. 2023	**Stabilizing and destabililizing mechanisms in fluid flows**, Hurghada, Egypt

Talks at University Seminars and Colloquia

2026	University of York (Mathematical Finance and Stochastic Analysis Seminar), University of Cambridge (Geometrical Analysis and PDE seminar)
2025	ENS Paris, Sorbonne University and University of Paris Cité (Nonlinear Analysis and PDE Seminar), University of Warwick (PDE and Applications seminar), King's College London (Analysis Seminar), University of Exeter (Geophysical and Astrophysical Fluid Dynamics seminar), University of Pisa (Seminars in Probability, Stochastic Analysis and Statistics), University of Victoria (Applied math seminar), Duke University (Analysis Seminar)
2024	Princeton University (Analysis Seminar), Courant Institute NYU (Analysis Seminar), University of Surrey (Mathematics Colloquium), Institut Henri-Poincaré (Paris-London analysis seminar), University of Copenhagen (Øresund Seminar on Analysis, PDE & Related Topics), University of Rome "Tor Vergata" (Seminario di Equazioni Differenziali), University of Bath (Analysis seminar), University of Warwick (PDE and Applications seminar)
2023	CNR-IAC Rome (Seminario Volterra), Bocconi University (Analysis Seminar), UT Austin (Analysis Seminar), Institute of Advanced Studies (Analysis & Mathematical Physics Seminar), Queen Mary University of London (Geometry, Analysis and Gravitation Seminar), UC Davis (PDE and Applied Math Seminar)
2022	AIMS-Senegal (PDE seminar), ICMAT (PDE seminar), SISSA (Analysis Seminar), University of Cambridge (PDE Seminar)
2021	Karlsruhe Institute of Technology (PDE Seminar), Mathematical Institute of the Czech Academy of Sciences (Necas PDE seminar), Université Claude Bernard Lyon 1 (Séminaire MACS), Gran Sasso Science Institute (Colloquium)
2020	Princeton University (Analysis of Fluids and Related Topics Seminar), University of Minnesota (PDE Seminar), Brown University (PDE Seminar), Old Dominion University (Colloquium), Stanford University (Applied Math Seminar), University of Bath (Analysis Seminar), MIT (PDE/Analysis Seminar), NYU Abu Dhabi (Colloquium), Durham University (Analysis & PDE seminar), SISSA (Analysis Seminar)

2019 Courant Institute NYU (Analysis Seminar), University of Maryland (PDE-Applied Math Seminar), Universidad de Sevilla (Fluid Conversations), ETH Zurich (Analysis Seminar), University of Warwick (PDEs and their applications Seminar)

2018 University of Basel (The BZ Seminar), University of Cambridge (Geometrical Analysis and PDE seminar), EPFL (Analysis Seminar), Imperial College London (Fluid Dynamics Seminar), University of Oxford (PDE CDT Lunchtime Seminar), UC San Diego (Analysis Seminar)

2017 Imperial College London (Applied PDEs Seminar), University of Utah (Colloquium)

2016 University of Miami (Colloquium), University of Maryland (PDE-Applied Math Seminar), USC (CAMS Colloquium), University of Virginia (Harmonic Analysis & PDEs Seminar), Tulane University (Probability and Statistics Seminar), CUNY (Non-Linear Analysis Seminar), Princeton University (Analysis of Fluids and Related Topics Seminar)

2015 University of Maryland–Baltimore County (Applied Math Colloquium), United States Naval Academy (Applied Math Seminar), Brown University (Analysis Seminar), Virginia Tech (Colloquium), University of Maryland (CSCAMM seminar)

2013 Indiana University (PDE/Applied Math Seminar), Politecnico di Milano (Mathematics Seminar), University of South Florida (Colloquium)

2010 Indiana University (PDE/Applied Math Seminar)

Talks at Conferences

2027 Dynamics across Scales: Bridging Kinetic Theory and Fluid Mechanics (ESI Vienna, Austria)

2026 Evolution Equations and Dynamical Systems (Politecnico di Milano, Italy), Recent advances on transport-diffusion PDEs and their applications (Università di Firenze, Italy), Fluids-waves-and -sand-beach (Cagliari, Italy), New Perspectives on Evolution Equations and Related Topics (Queen Mary University of London), Singularities, non-uniqueness and complex dynamics of incompressible fluids (University of Bath)

2025 Beyond the Horizon: Navigating Turbulence, irregularity, and Stochasticity in Fluid Dynamics (ICMS, Edinburgh), Deterministic and Stochastic Advection in Fluid Dynamics (University of Basel), Coherent structures and (in)stabilities in fluids (GSSI, Italy), Kinetic theory and fluid mechanics: couplings, scalings and asymptotics (CIRM, France), Clifford Lectures (Tulane, New Orleans)

2024 PDE Festival@UniMi (Milan, Italy), Discontinuities and Singularities of Solutions to Nonlinear Evolution PDEs (Oxford, UK), Trends and perspectives in the analysis of PDEs (Rome, Italy), European Congress of Mathematics (Seville, Spain), Alhambra PDE Days (Granada, Spain), Equadiff 2024 (Karlstad University, Sweden), Wave Dynamics and Fluid-Structure Interactions (Lake Como School of Advanced Studies, Italy), Stochastic dynamics and stochastic equations (EPFL), Fluids in Seoul (HCMC, Seoul)

2023 Journée de Contact FNRS (Université Catholique de Louvain), Young researchers in PDEs (ICMAT, Madrid), Free Boundary Problems (Columbia University), Nonlinear Evolution PDEs, Fluid Dynamics and Transport Equations (Erice, Italy), KineticNet (online seminar), Multiscale Analysis and Methods for Quantum and Kinetic Problems (IMS, National University of Singapore)

2022 When Kinetic Theory meets Fluid Mechanics (ETH Zurich), Small scale dynamics in fluid motion (Simons Center for Geometry and Physics, Stony Brook), Long Time Behavior and Singularity Formation in PDEs-Part V (NYU Abu Dhabi), CY Days in Nonlinear Analysis (Maison internationale de la recherche, France), Modelling and analysis of turbulent transport, mixing and scaling (Newton Institute, UK), Transport, Fluids and Mixing (Centro di Ricerca Matematica Ennio De Giorgi, Pisa)

2021	12th Forum of Partial Differential Equations (Będlewo, Poland), New Trends in Nonlinear Diffusion: a Bridge between PDEs, Analysis and Geometry (Casa Matemática Oaxaca, Mexico), 8th European Congress of Mathematics (Portorož, Slovenia), SIAM Conference on Applications of Dynamical Systems, Oxbridge PDE Conference
2020	Partial differential equations describing far-from-equilibrium open systems (Charles University, Prague), Workshop on Euler and Navier-Stokes Equations (Fields Institute, Toronto), AMS Fall Eastern Sectional I Meeting, PDE Seminar via Zoom, Winter school on Turbulence in PDEs (EPFL)
2019	SIAM Conference on Analysis of PDEs (La Quinta, CA), Dynamics, Equations and Applications (University of Science and Technology, Kraków), PDE 2019: Partial Differential Equations in Fluids and Solids (WIAS, Berlin), Mathematical Aspects of Hydrodynamics (MFO, Oberwolfach), Summer School in Analysis of PDEs and Fluid Dynamics (ICMS, Edinburgh), GMT and PDEs (University of Basel), XVIII Italian Meeting on Hyperbolic Equations (Palermo), Transport, Mixing and Fluids (Westfälische Wilhelms-Universität Münster)
2018	Nonlinear PDEs from the Oceanic and Atmospheric Dynamics (CAMIS, Guangzhou, China), The 12th AIMS Conference (Taipei, Taiwan), An Analyst, a Geometer and a Probabilist Walk Into a Bar (Cardiff University), Mathematical fine structures in fluid dynamics (GSSI, Italy)
2017	SIAM Conference on Analysis of PDEs (Baltimore), Mathematical Congress of the Americas (Montreal), AMS Eastern Sectional Meeting (CUNY, New York), Selected topics in transport phenomena (University of Maryland), Joint Mathematics Meetings (Atlanta)
2016	Optimal Control for Evolutionary PDEs and Related Topics (Cortona, Italy), International Conference on Evolution Equations (Vanderbilt University), AIM Workshop on Mixing and nonlinear stability (San Jose, CA), AMS Southeastern Sectional Meeting (University of Georgia), Joint Mathematics Meetings (Seattle, WA)
2015	SIAM Conference on Analysis of PDEs (Scottsdale, AZ), Workshop on Stochastic PDEs (University of Pittsburgh), KI-net Young Researchers Workshop (CSCAMM, University of Maryland), AMS Central Sectional Meeting (Loyola University, Chicago), Mathematics of Geophysical Flows and Turbulence (Fudan University, Shanghai, China), EquaDiff 2015 (Université Claude Bernard Lyon 1, France)
2014	AMS Western Sectional Meeting (San Francisco State University), The 10th AIMS Conference (Madrid, Spain), AMS Southeastern Sectional Meeting (University of Tennessee, Knoxville), Joint Mathematics Meetings (Baltimore, MD)
2013	SIAM Conference on Analysis of PDEs (Lake Buena Vista, FL), AMS Central Sectional Meeting (Washington University, St. Louis, MO), AMS Southeastern Sectional Meeting (University of Louisville)
2012	The 9th AIMS Conference (Orlando, FL), SIAM Southeastern Atlantic Section Conference (University of Alabama, Huntsville)
2010	AMS Central Sectional Meeting (University of Notre Dame)
2009	Mathematical models and analytical problems for special materials (Università di Brescia)

Professional service

Nov. 2026	Co-organizer of the Oberwolfach Seminar "Mathematical Developments in Magnetohydrodynamics and Dynamo Theory", MFO, Oberwolfach, Germany
Aug. 2026	Co-organizer of the workshop "Stable and unstable Dynamics in PDEs and Hamiltonian Systems", Casa Matemática Oaxaca, Mexico
Jun. 2026	Co-organizer of the conference "Turbulent days", Imperial College London

Jun. 2025	Co-organizer of the workshop "Blurring the lines between pure and applied through mixing", Sabhal Mòr Ostaig, Isle of Skye, Scotland
Jun. 2024	Co-organizer of the mini-symposium "Recent developments in mathematical fluid dynamics", Equadiff 2024, Karlstad University, Sweden
Feb. 2024	Co-organizer of the conference "Phase mixing, kinetic theory and fluid mechanics", SwissMAP Research Station, Les Diablerets, Switzerland
Oct. 2023	Co-organizer of the conference "Mathematical Aspects of Geophysical Flows", IAC-CNR, Rome, Italy
Jul. 2023	Co-organizer of the summer school and conference "Stability and dynamics in fluid mechanics and kinetic theory", Imperial College London
2023–	ERCOFTAC, Special Interest Group on Mathematical Fluid Mechanics, steering committee
2022–	London Mathematical Society, membership
2022–	Gran Sasso Science Institute, governing board
2020–	EPSRC Associate College Member
2017–	Co-organizer of the Applied PDEs Seminar, Imperial College, London
Dec. 2019	Co-organizer of the special session "Mixing and Stability in Fluids", SIAM Conference on Analysis of Partial Differential Equations, La Quinta, CA
Oct. 2018	Co-organizer of the special session "Analytical and Numerical Aspects of Turbulent Transport", AMS Fall Central Sectional Meeting, University of Michigan, Ann Arbor, MI
Oct. 2016	Co-organizer of the special session "Nonlinear and Stochastic Partial Differential Equations", AMS Fall Western Sectional Meeting, University of Denver, Denver, CO
Dec. 2015	Co-organizer of the special session "Deterministic and Stochastic Aspects of Fluid Dynamics", SIAM Conference on Analysis of Partial Differential Equations, Scottsdale, AZ
Nov. 2015	Co-organizer of the KI-net Young Researchers Workshop "Kinetic theory with applications in physical sciences", CSCAMM, University of Maryland, College Park, MD
Aug. 2015	TA at the workshop "Incompressible Fluid Flows at High Reynolds Number", Mathematical Sciences Research Institute, Berkeley, CA
Mar. 2015	Co-organizer of the special session "Mathematical fluid dynamics and turbulence", AMS Spring Eastern Sectional Meeting, Georgetown University, Washington, DC

Teaching Experience

2022	**AIMS Senegal** - Applied PDEs (Modelling)
2017–	**Imperial College London** - Advanced PDEs 2, Theory of PDEs, Introduction to PDEs, TCC Introduction to Euler equations of incompressible fluids, Stochastic Differential Equations, MPC1 Maths and Physics for Chemists I
2014–2017	**University of Maryland** - MATH130/MATH131 Calculus I & II for the Life Sciences, MATH410/MATH411 Advanced Calculus I & II
2012–2013	**Indiana University** - D117/D118 Introduction to finite math I & II, M118 Finite Mathematics, M027 Precalculus with trigonometry